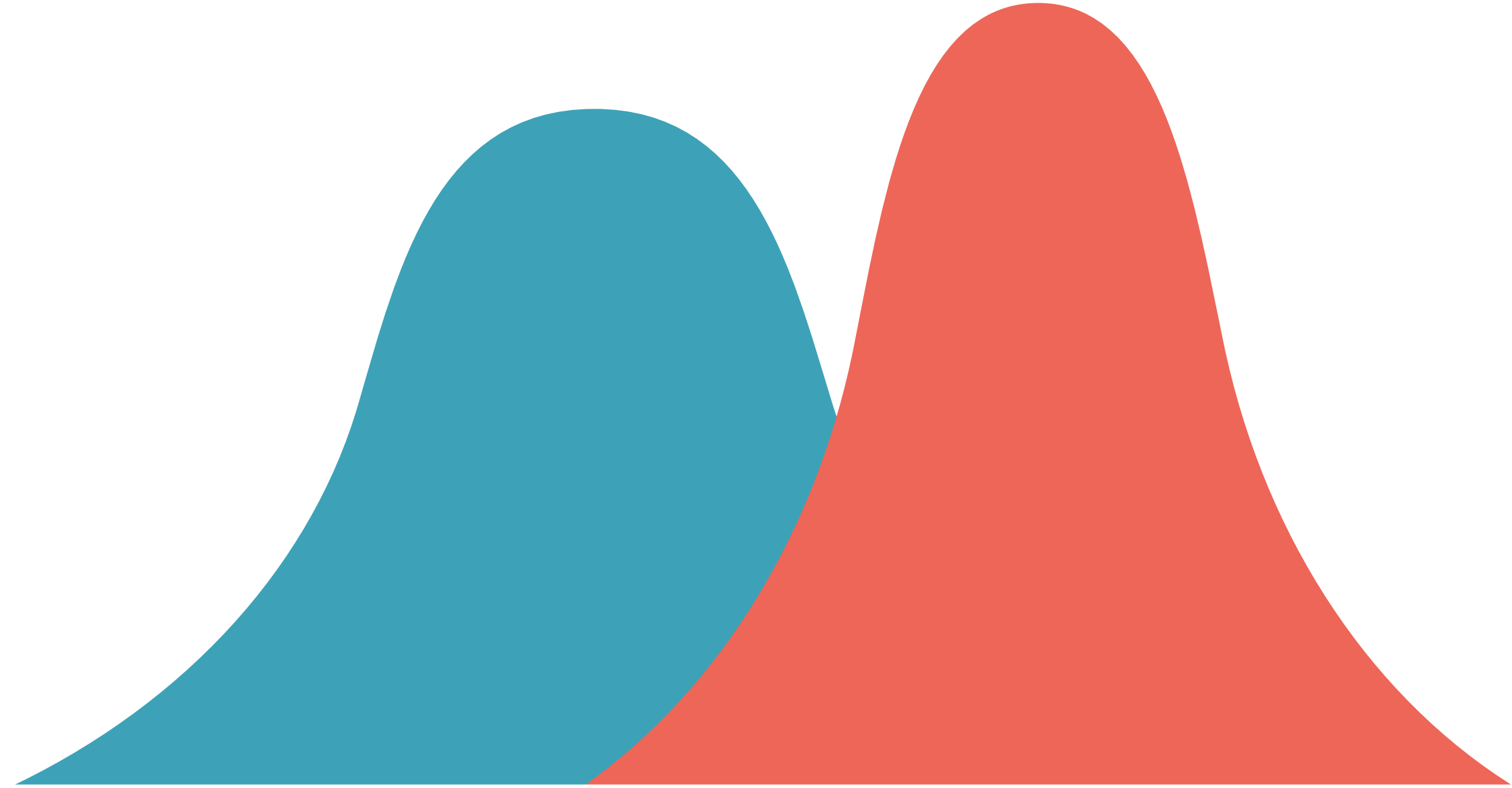
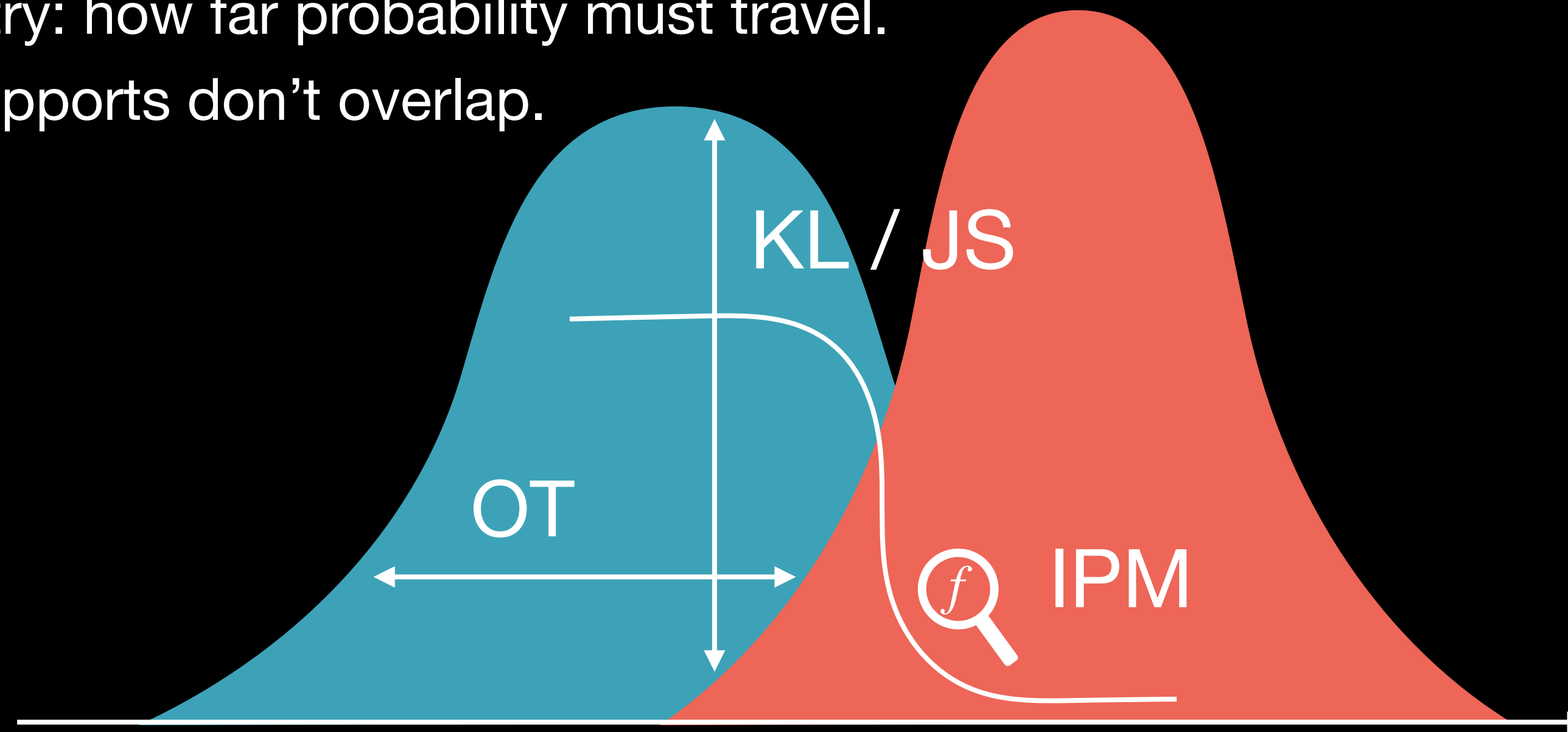




Three Ways to Compare

1. “Horizontal” (OT / Wasserstein)

- Compare distributions globally by *moving* mass.
- Sensitive to geometry: how far probability must travel.
- Finite even when supports don't overlap.



2. “Vertical” (f-divergences: KL, JS, etc)

- Compare locally via *pointwise* ratios of densities.
- Sensitive to overlap: ∞ if supports don't match.
- Natural for likelihoods / information-theory

3. “Functional” (IPMs: MMD, etc)

- Compare via expectations of a test function class.
- Witness function highlights where distributions differ.
- Smoother than OT, more global than KL.

Formal Setup